

Constructing MAC: Three Approaches

Design ‘from scratch’, validate security by failure to cryptanalyze

- Huge effort, risk do only for a few ‘building blocks’
- Maybe from EDC (Error Detection Code), but it is not secure for every EDC

Robust combiner of (two) MAC candidates:

$MAC_{k,k'}(m) = f_k(m) || f_{k'}(m)$, $MAC_{k,k'}(m) = f_k(m) \oplus f_{k'}(m)$ are secure MAC, if *either* f or f' are a secure MAC

Every PRF is also a MAC

- A PRF is a MAC for l -bit messages.
- (l, n) -bit FIL (Fixed Input Length) MAC from n -bit PRP (block cipher): use CBC-MAC

Cipher Block Chaining MAC: CBC-MAC

Split plaintext m into blocks

Fixed, known (zero) Initialization Vector (IV)

The tag is the cipher of the last block $CBC - MAC_k^E(m_1 || m_2 || \dots || m_l) = E_k(m_l \oplus E_k(\dots E_k(m_1)))$

Recall: MACs are deterministic functions

CBC-MAC-based VIL-MAC

Is $CBC - MAC^E$ a VIL-MAC?

- No!
- Ask for $b = CBC - MAC_k^E(a) = E_k(a)$;
- then output (ac, b) s.t $m = ac$ with $tag = b$ where $c = a \oplus b$.

Solution: prepend message length (called CMAC)

- Let $CMAC_k^E(m) = CBC - MAC_k^E(L(m) || m)$, where $L(m)$ is a 1-block encoding of $|m|$
- CMAC is a secure VIL-MAC construction!
- First block is always the message length. Check rest of tag based on MAC and msg length

Examples of MAC Constructions

Are the following constructions a secure MAC?

1. Let E_k be a block cipher that takes input of length n bits. For a msg m of length $2n$ bits, compute the tag as $MAC_k(m) = E_k(m_L) \oplus E_k(m_R)$
Function is deterministic
Ex: $m = 0^n || 1^n$, get tag t . $m' = 1^n || 0^n$, tag is equal to t .
Attacker probability: $P(\text{Valid Forgery}) = 1 ; 1 - \frac{1}{2^n}$
2. Let G be a secure PRG. For a msg m of length n bits, compute the tag as $MAC_k(m) = k \oplus PRG(m)$
Attacker can determine $y = G(m)$, and get k via $MAC_k(m) \oplus y = k$
 $\Sigma = 1 - \frac{1}{2^n}$

Combining Authentication and Encryption

For confidentiality, use encryption

For authentication, use MAC

For both confidentiality and authentication?

- Option 1: Combine MAC and encryption
Potential pitfalls (vulnerabilities)
- Option 2: authenticated-encryption schemes (or modes)
Easier to deploy (securely)

Generic MAC and Encryption Combinations

Three standards, three ways...

- Authenticate and Encrypt (A&E)
- Authenticate then Encrypt (AtE)
- Encrypt then Authenticate (EtA)

Security of Generic MAC/Enc Combinations

A&E may be vulnerable!

- Example:
Never compute a MAC over the information directly! Only for integrity, not security

How about EtA? **Provably CCA-Secure!!**

- wow!

Keys for MAC and Encryption?

Using same key for MAC+Encryption? Never!

Exercise: show (contrived) examples' vulnerabilities:

Solution: $k_{mac} = PRF_k('MAC')$, $k_{enc} = PRF_k('Encrypt')$

Attacker does not know k !